

REPORT

Exploratory Data Analysis (EDA) Project

Submitted by:

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Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset to understand data patterns, relationships, and trends through statistical analysis and visualization.

Dataset

Titanic Dataset

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook
- Scikit-Learn

Dataset Information

```
[3]: print(df.info())

<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   PassengerId     891 non-null    int64  
1   Survived        891 non-null    int64  
2   Pclass         891 non-null    int64  
3   Name            891 non-null    str     
4   Sex             891 non-null    str     
5   Age            714 non-null    float64 
6   SibSp          891 non-null    int64  
7   Parch          891 non-null    int64  
8   Ticket         891 non-null    str     
9   Fare           891 non-null    float64 
10  Cabin          204 non-null    str     
11  Embarked       889 non-null    str     
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
None
```

Figure 1: Dataset Structure and Information

Explanation:

This figure provides information about the dataset, including the number of rows, columns, data types, and missing values. Understanding the dataset structure is an essential first step in exploratory data analysis.

Statistical Summary

```
[4]: print(df.describe())
```

	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

Figure 2: Statistical Summary of Dataset

Explanation:

This statistical summary provides key measures such as mean, minimum, maximum, and standard deviation for numerical features. These statistics help understand the overall distribution of the data.

Correlation Heatmap

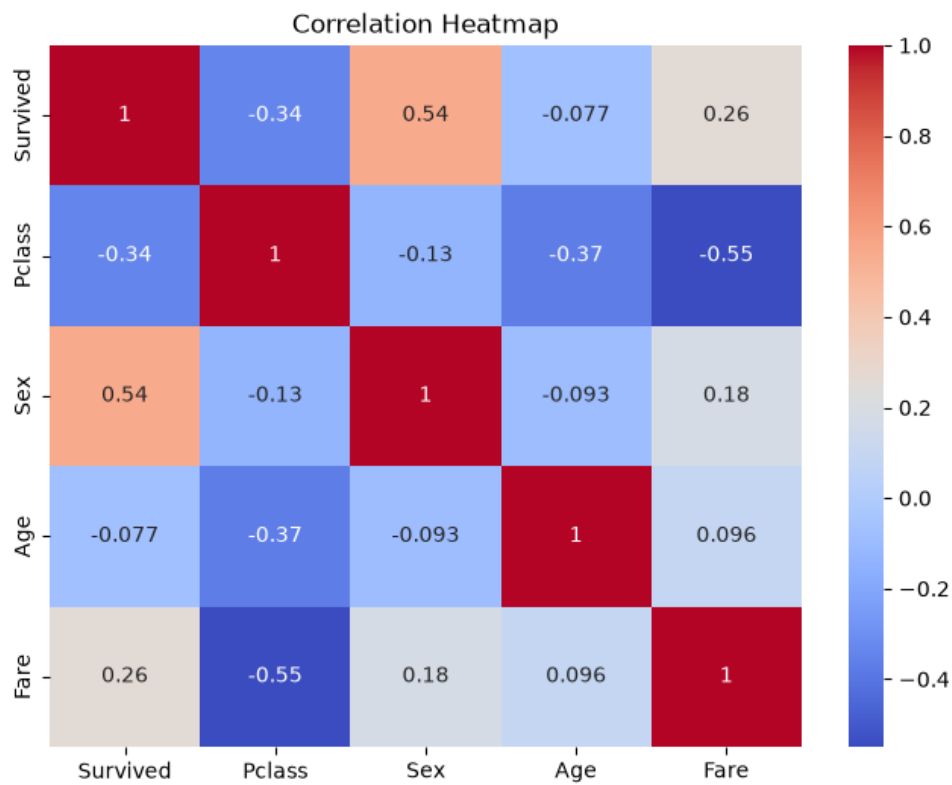


Figure 3: Correlation Heatmap

Explanation

The correlation heatmap visualizes the relationships between numerical variables in the dataset. Positive and negative correlations help identify factors that may influence passenger survival.

Observation

- Gender and passenger class show noticeable relationships with survival.
- Some features have stronger influence on survival outcomes than others.

Age Distribution Analysis

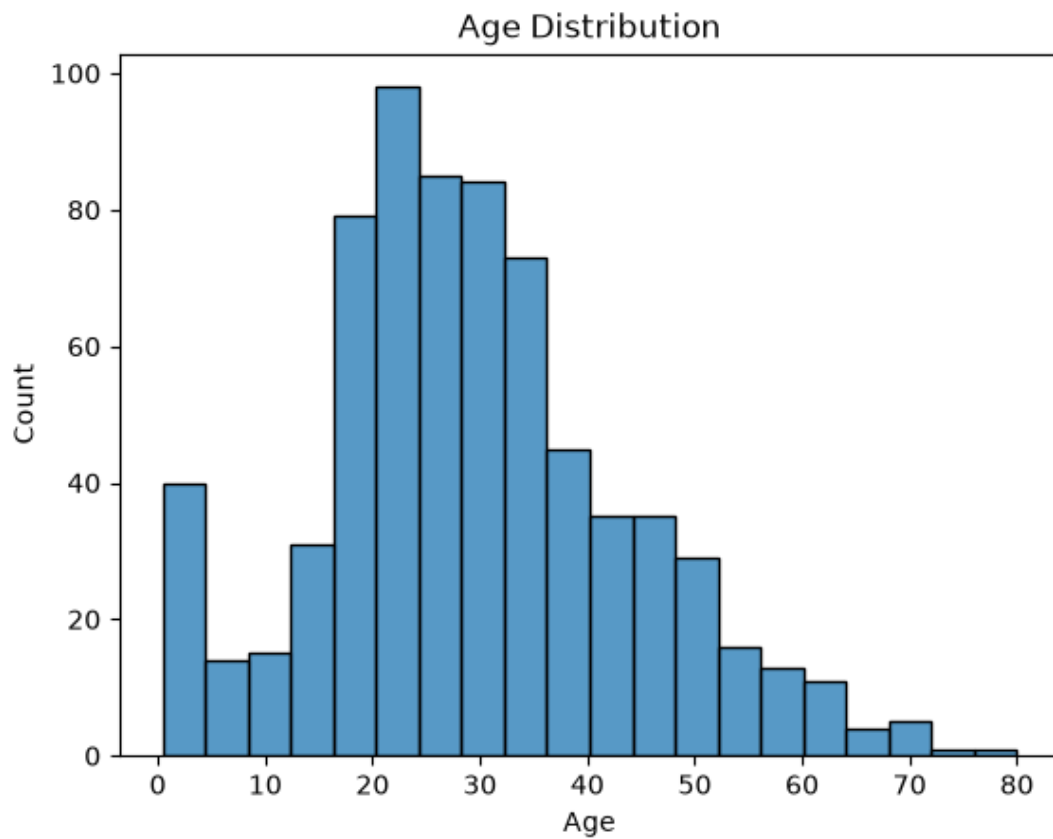


Figure 4: Age Distribution Analysis

Explanation

This histogram shows the distribution of passenger ages and helps identify common age groups within the dataset.

Observation

- Most passengers were between 20 and 40 years old.
- The dataset contains passengers across a wide age range.

Survival by Passenger Class

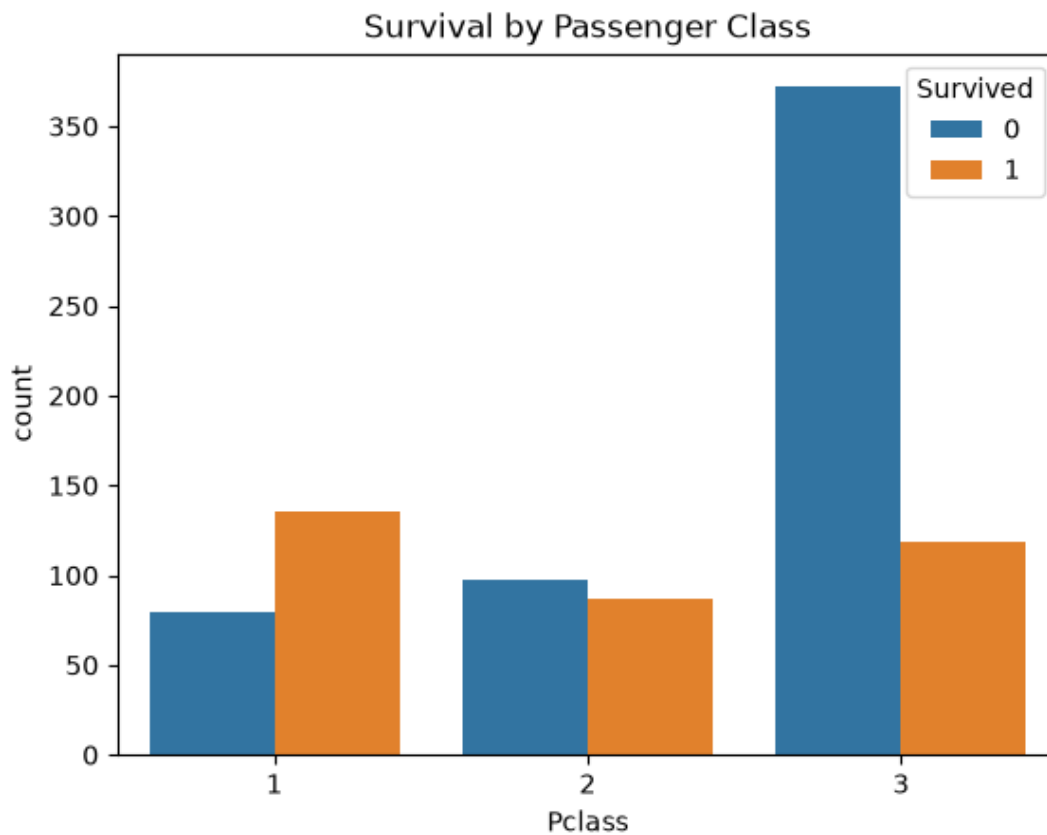


Figure 5: survival by class

Explanation

This chart compares passenger survival across different ticket classes.

Observation

- First-class passengers had a higher survival rate.
- Third-class passengers experienced lower survival rates.

Gender Distribution

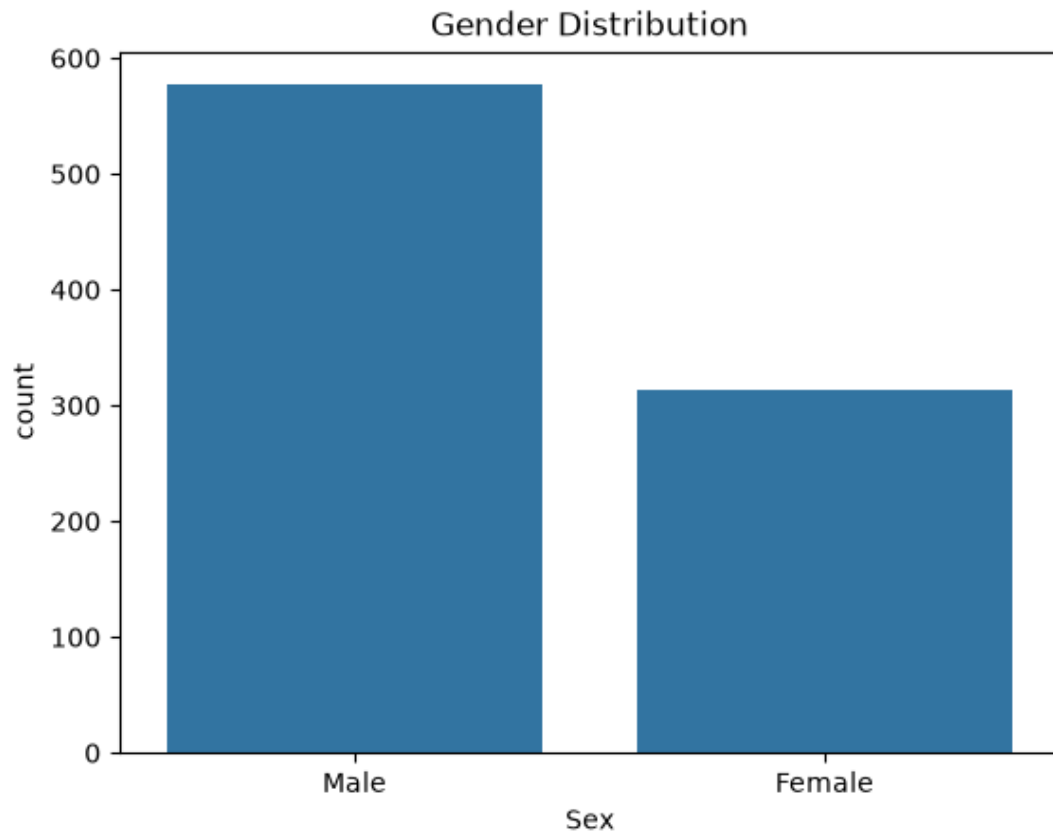


Figure 6: gender distribution

Explanation

This chart displays the number of male and female passengers in the dataset.

Observation

- Male passengers were more numerous than female passengers.
- Gender distribution helps provide context for further survival analysis.

Conclusion

Key Findings

1. The Titanic dataset contains valuable demographic and passenger information.
2. Passenger class significantly influenced survival rates.
3. Age distribution was concentrated among young and middle-aged passengers.
4. Gender played an important role in survival outcomes.
5. Correlation analysis revealed relationships among key variables.

Conclusion

This Exploratory Data Analysis project successfully examined the Titanic dataset through statistical summaries and visualizations. The analysis provided meaningful insights into passenger demographics, survival patterns, and feature relationships. EDA serves as an important step before applying machine learning techniques.